

NGO "Kosmos Tabir"
innovation humanitarian solution

Mini Solar Power System for de-occupied areas in Ukraine

"Mini Solar Power Plants" is an innovative solution developed by NGO "Kosmos Tabir".

The main idea and goal is to equip all residential houses and administrative buildings with mini solar power stations, allowing devices to be charged for free, even without sunlight, at 5-12V. This includes smartphones, tablets, Wi-Fi routers, flashlights, power banks, construction tools, walkie-talkies, and other equipment



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Problematic:

In Ukraine, as a result of ongoing combat, over 475 settlements* have been left without power, with most remaining **continuously without electricity**. This affects tens of thousands of households in the Kherson, Zaporizhzhia, Donetsk, Kharkiv, Sumy, and Chernihiv regions. People cannot contact their loved ones or doctors, and children are unable to receive an education.

This situation presents a significant challenge to the humanitarian sector as a consequence of modern warfare, with the potential for similar situations to spread globally.

For example, in the Kherson region, more than 26,000 households* remain consistently without electricity, with the majority having been without power for nearly three years.

[*Data from the Ministry of Energy of Ukraine - July 2024](#)



Kherson region

Up to 26,000 households with no electricity for almost 3 years

*Data from the Ministry of Energy of Ukraine - July 2024



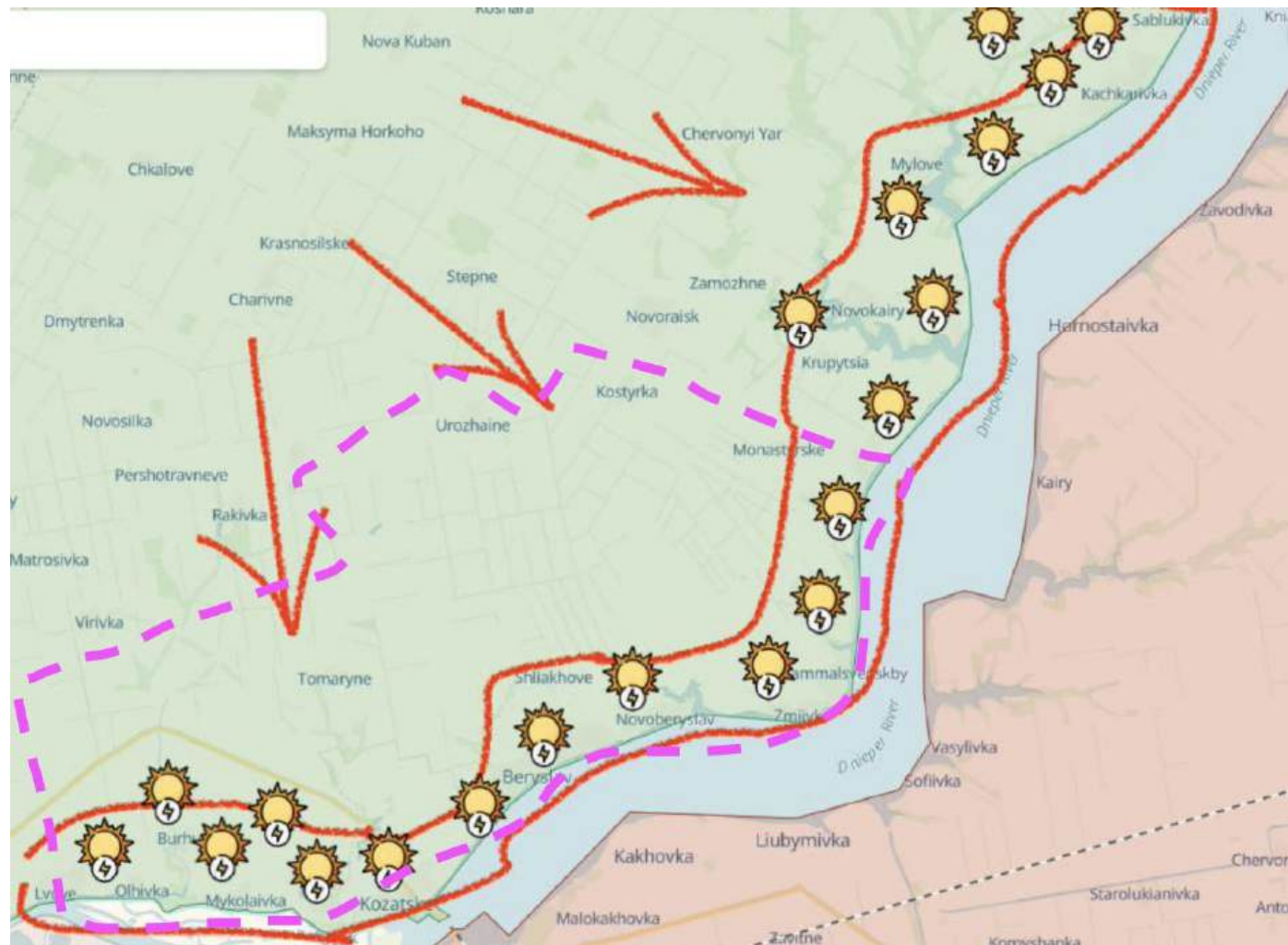
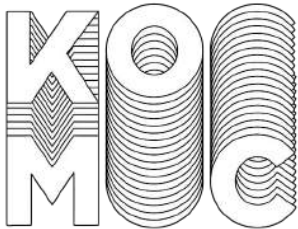
At NGO “Kosmos Tabir” in the past year we have made 4 humanitarian missions to Kherson region, mostly Beryslav area — critical area which are shelled daily and have no electrical power



Installed in the
Berislav district:

30 small (0.5 kWt)
batteries + solar
station - one per
village

400 Mini Solar
Power Plants - one
per household



400 Mini Solar
Power Plants -
one per
household



[Source in Facebook](#)



Just two week after we delivered our mini solar system, community center in the **village of Beryslav area** was hit by shrapnel from russian guided bomb that exploded near it

Our system still work fine, even **after bomb explosion near**, solar panel was hit by shrapnels, but still works and charging phones and building instruments for fast renovation

So **our system is stable and efficient near the front line**, that's why we call it an innovation humanitarian solution



Source:
[Web page](#)
[Kherson region](#)
[administration](#)

For more than a year installing and testing
it, we finally created innovative solar
solution that we plan to give to every living
household that was affected by war and
needs electricity

What is mini Solar Power System and it works ?

Mini solar power system is built in similar to MPPT technology. It has inductor, capacitors, and switching components (like MOSFET). These converters adjust the input voltage to a desired output voltage efficiently. **Difference that our system stabilise voltage for 5v/12v and 1-2A.**

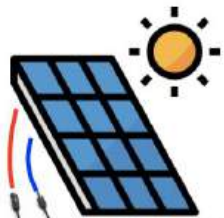
It have 4 USB outputs, 3 for 5v, 1-2A and 1 for 9-12v (automatic identification)

It goes **with 10 meters outdoor solar** (analog) cable 2x0.75mm with MC4 solar connectors

It **work with any solar panel** - used, cracked, and even with gunshots or shrapnel damage
*(with voltage range 18v-60v of solar panel)

System is extremely simple and easy to install - understandable for the elderly as well as for children

Mini Solar Power System cost without panel is around 10-12€ which makes it simple and super useful solution with possibility to instal it **on every living household near frontline** and postwar (de-occupied) areas



System consume 60-80 Watt
Possibility to plug
1-3 systems to 1 panel

12v instrument
WiFi Router
Walkietalkie
Shaver

SOLAR
MINI POWER
SYSTEM



12V

5v

Extra cables for
one more Mini Solar System

Storage energy
in a regular
powerbank

FM

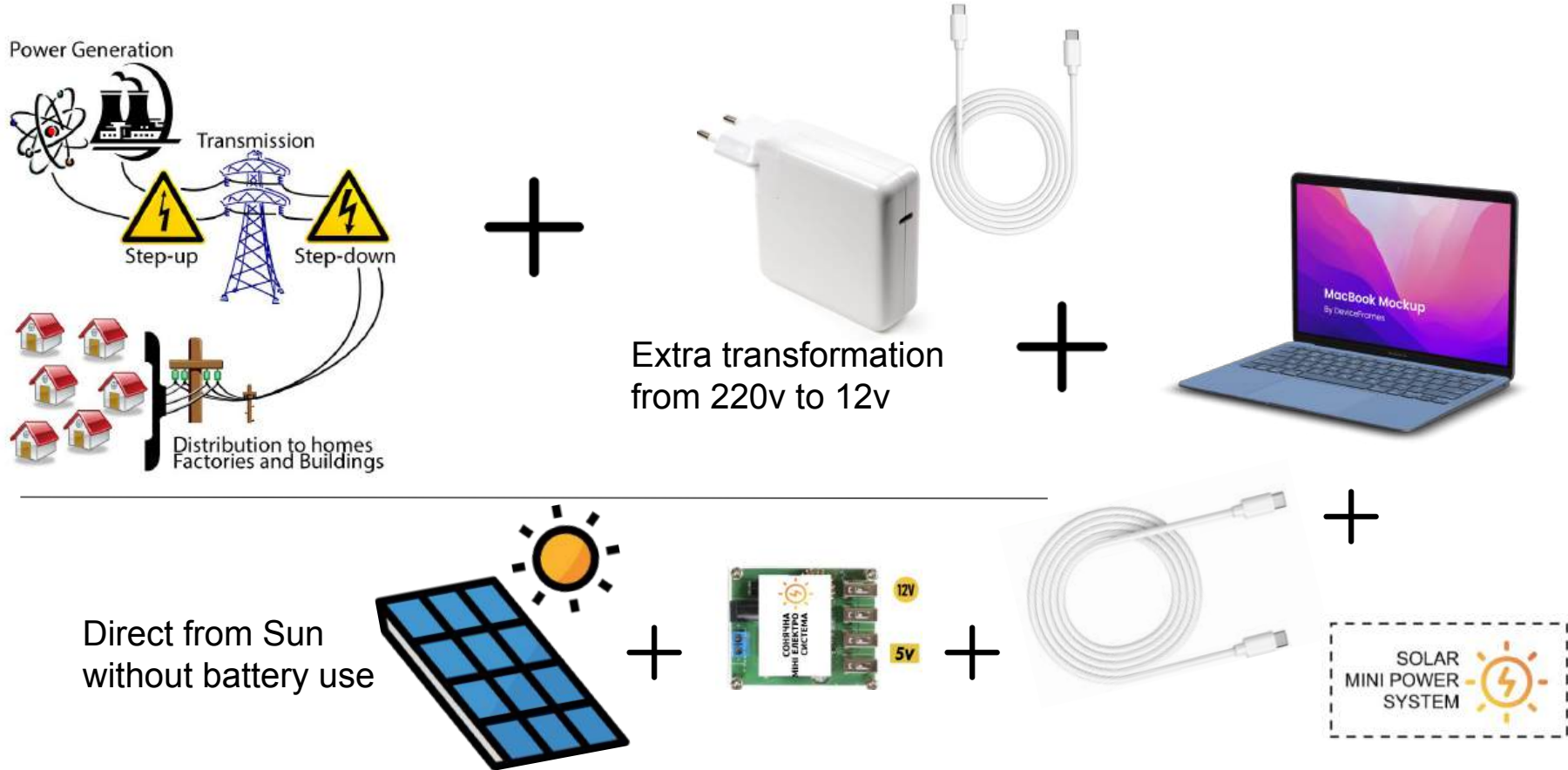
*WiFi Router works from
powerbank with 12v DC cable

Any type of
USB lights

Smartphone
Laptop
Tablet

Portable shower
Water Pump Wash

What better to charge your laptop?



Example - Test Gunshots solar panel

Trina Solar TSM-445DE17M(II) 445w 49v

*damage by shrapnel from russian tank and gunshots
near Kyiv Area in 2022

Works fine, have more than 50% from the
starting efficiency

Tested for 4 month, system with
gunshots panel works fine, even with
cloudy weather

In full power, system consume just 60-80
watt.

For this case can be two system per one
specific solar panel

Same thing with used solar panel



Solar Panels Face Recycling Challenge in Europe

1. Lack of Standardization in Solar Panel Composition
2. Expensive to recycle, cheaper to keep in storage
3. Environmental and Health Risks of Current recycling technology
4. Regulatory Hurdles and Lack of Awareness

European study shows continent exports solar waste outside of EU

Because it's not waste...

... for such areas as de-occupied regions in Ukraine, areas near the front line.

It's only one option to charge phone and to call relatives in many villages in Ukraine now

Let's give them a second life and let's give electricity to people for Humanitarian purpose



[Pic source bbc.com](https://www.bbc.com)

It's happening
anyway, let's give
this issue
reasonable and
justified
humanitarian
purpose

European study shows continent exports solar waste

Europe's solar boom is hiding its waste through exports, according to a new study. Instead of recycling, as expected under European regulations, vast amounts of solar panel waste are exported, undermining sustainability goals. The researchers suggest a digitized value chain as a potential fix to work toward a circular economy.

APRIL 16, 2024 **TRISTAN RAYNER**

MODULES & UPSTREAM MANUFACTURING

POLICY

TECHNOLOGY AND R&D

EUROPE

WORLD



SOLAR
MINI POWER
SYSTEM



[Source pv-magazine](https://www.pv-magazine.com)

We don't need new panels, we need used PV
for living households near the frontline

About us

Basically we are a creative community that transforms to humanitarian innovation projects

Meet us by [pics](#) and [video](#) tour

Photo credit:
NGO “Kosmos Tabir” community in culture trip to Herson region in 2020 by ГогольTrain
Now all this area are at war and occupation
[First Art train in Europe - video from Mariupol](#)



Our partners on this humanitarian mission

Institutional partners



Ministry
of Energy of Ukraine



Office of the
President of Ukraine



Kherson regional
state administration



Kharkiv regional
state administration

Project partners



What we need for humanitarian mission

We are looking for partnership with strong institutions, associations, NGOs, solar plants and companies - to collect used PV and solar equipment for Ukraine



Used Solar Panels

Not less than 200w and no less than 50% efficiency left.
Can be cracked or lightly damaged



Used Vans or Trucks

Can be old or cosmetic damage, but working



Used Inverters and batteries

LiFePo4 batteries only, and not less than 60% efficiency left.
Used solar inverters.

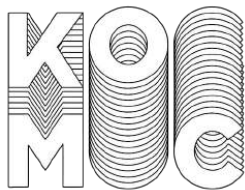


Used Solar Equipment

Cables, fastening, crimpers

Also looking for Project partners to provide funding to living households for our system installations





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areas in Ukraine**



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